UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFORMATION

Clear Form | Copy Sheet | Update Index

School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSE3103**	**Academic Session/Semester:**	**2026 2027 1**
Course name:	**Cognitive Computing**	**Pre/co requisite (course name and code, if applicable):**	
Credit hours:	**3**		

Course synopsis	This course provides an intermediate-to-advanced introduction to Cognitive Computing, tailored for software engineering students to explore how computational systems emulate human perception, reasoning, and decision-making. It bridges artificial intelligence principles with practical software engineering patterns required to build intelligent applications. The course covers foundational cognitive architectures, knowledge representation models, probabilistic reasoning engines and natural language processing. Taught through interactive lectures, hands-on coding labs, and project-based learning, the course emphasizes systems thinking, team collaboration, and practical software design. By the end of the course, students will be able to analyze cognitive architectures, design cognitive applications, and collaborate effectively to construct secure, ethical, and trustworthy cognitive software prototype.			
Course coordinator (if applicable)	Dr Norsham binti Idris			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr Norsham binti Idris			norsham
	Dr. Sim Hiew Moi			hiewmoi

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	[1]CLO	PLO [2](MQF Cluster Code)	[3]Taxonomies & [4]Graduate attributes	[5]ESD	[6]VBE	T&L methods	[7]Assessment methods
CLO1	Design cognitive architectures, knowledge models, and non-deterministic systems for software solutions.	PLO2 (C2)	C4 (SC2)			Lecture, active learning	Quiz (10%), Test (10%), Final (20%) Project Asg (5%)
CLO2	Develop cognitive applications using a range of intelligent methods to solve a domain problem.	PLO3 (C3A)	C6 (SC3)	STT		Project-based learning	Quiz (5%), Test (5%), Final (20%), Project Asg (10%)
CLO3	Collaborate and implement trustworthy and ethical cognitive systems in team-based software engineering workflows.	PLO4 (C3B)	C5 (SI1)		Synergy	Project-based learning	Project Asg (10%), Pr(5%)

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.

[1]**Up to 5 CLO**

[2]**MQF Cluster Code**

C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

[3]**Taxonomies of Learning Domain** and [4]**UTM's Graduate Attributes** for measurement of outcomes achievement

[5]**Education for Sustainable Development (ESD)**

Systems Thinking (ST); Anticipatory Thinking (AT); Normative (N); Strategic Thinking (STT); Collaboration (C); Critical Thinking (CT); Self-Awareness (SA; Integrated Problem Solving (IPS)

[7]**Value-Based Education (VBE)**

Integrity; Synergy; Excellence; Sustainability

[7]T – Test; Q – Quiz; HW – Homework; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc. Please specify only the related type of assessment.

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities
2	Project-based learning	Conducted through design assignments. Students in a group of 3 are given 2 design projects that require power electronics solutions involving the design calculations and verification using MATLAB/Simulink. Compliance to the design specifications need to be given in the form of written reports.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

Cognitive Skills, Practical Skills, Collaborative & teamworking.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory ☒	Learning and Teaching Activities									TOTAL SLT
			Face-to-Face (F2F)							Non F2F Independent Learning		
			Physical			Online (Synchronous)				Online (Asynchronous)	Others	
			L	P	O	L	T	P	O			
Week 1 *12 -16 Oct 2026*	**Chapter 1 : Computer and Intelligence** Introduction to thinking and intelligence, Responsible AI, Computational Intelligence, AI Applications.	CLO1				3					2	5
Week 2 & Week 3 *19 -30 Oct 2026*	**Chapter 2: Knowledge, Reasoning & Decision Making** Knowledge Representation, Rule-based systems Inference, Search algorithms Introduction to fuzzy logic	CLO1	6	2						1	2	11
Week 4 & Week 5 *2 - 13 Nov 2026* 8-9 Nov-Deepavali	**Chapter 3: Machine Learning Fundamentals** Overview of machine learning, Supervised vs unsupervised learning, Classification, clustering, reinforcement, and regression, Model evaluation	CLO1	6	2						1	2	11
Week 6 *16 -20 Nov 2026*	**Chapter 4 : Cognitive Computing Concepts and Applications** Fundamental concepts of cognitive computing and its applications Basic cognitive system architecture	CLO1				3				1	2	6
Week 7 *23 -27 Nov 2026*	**Chapter 5: Artificial Neural Network** Overview of Artificial Neural Network (ANN) Perceptron ANN Structure and Activation Functions	CLO2	3	1						1	2	7

Week 8 *30 Nov-4 Dec 2026*	MID-SEM BREAK											
Week 9 *7 -11 Dec 2026*	**Chapter 5: Artificial Neural Network (cont)** Multilayer Perceptron Backpropagation in ANN					3					2	5
Week 10 & Week 11 *14-25 Dec 2026* (25 Dec - Christmas)	**Chapter 6: Deep Learning and Deep Neural Network** Overview and Concepts of Deep Learning Overview of Deep Neural Network (DNN) DNN Structure and Functions Classification using DNN	CLO2	6	2						1	3	12
Week 12 & Week 13 *28 Dec 2026-8 Jan 2027*	**Chapter 7 : Convolutional Neural Network (CNN)** CNN Architecture Convolution Operation CNN Training Basic applications	CLO2	6	2						1	2	11
Week 14 *11-15 Jan 2027*	**Special topic : Data Mining with Deep Neural Network** Overview of Data Mining Use of DNN for Data Mining Applications in Sentiment Analysis	CLO2				3					2	5
Week 15 *18-22 Jan 2027*	**Project Presentation & Demonstration**	CLO3	3								3	6
										SUB-TOTAL SLT :		**86**

	Continous Assessment	**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Quiz 1	5	1			1	2
2	Quiz 2	5	1			1	2
3	Quiz 3	5	1			1	2
4	Test 1	15	2			3	5
5	Project Proposal	5	1			1	2
6	Project Design	10	3			1	4
7	Project Prototype	10	3			1	4
5	Project Presentation	5	3			1	4
					SUB-TOTAL SLT :		**25**

Assessment activities carried out during the semester to monitor student progress (e.g., quizzes, assignments, projects).

Final Assessment		%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	40	3		6		9
Assessment conducted at the end of the course to evaluate the achievement of overall learning outcomes (e.g., final exam, final project).					SUB-TOTAL SLT :		9
					SLT for Assessment:		34
					GRAND TOTAL SLT:		120

A	% SLT for F2F Physical Component	50.83
B	% SLT for Online & Independent Learning Component :	49.17
C	%SLT for Online Component:	20.00
D	% SLT for All Practical Component:	7.50
D1	% SLT for F2F Physical Practical Component:	7.50
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

Computer lab

References (include required and further readings, and should be the most current)

1.Velankar, M. R., Mahalle, P. N., & Shinde, G. R. (2024). *Cognitive computing for machine thinking* . Springer.
2.Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach (4th ed.)* . Pearson.

Other additional information (if applicable)

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

Disclaimer:

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ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:	**Certified by:**
Name: DR.NORSHAM BINTI IDRIS	**Name:** PM Dr Mohd Adham Isa
Signature: 	**Signature:** 
Date: 30-Aug-26	**Date:** 30-Aug-26